ORF 309 Midterm 1 — Preparation Strategy

Slides 01–09 | Through Section 2.3 | March 6, 2026

Based on analysis of **8 years** of past midterms (2017–2025) and all 9 slide decks

Topic Frequency Analysis

The table below shows how often each topic has appeared across all available past midterms. Use this to prioritize your study time.

Topic	**Appeared in**	**Priority**
Bayes' Theorem / LOTP	**every single exam**	Must-know
Counting / Combinatorics	**every exam**	Must-know
Expectation (linearity, indicators)	**every exam**	Must-know
Geometric distribution + series	7/8 exams	Must-know
Binomial distribution	6/8 exams	Must-know
Variance computation	5/8 exams	High
Independence (verify/use)	5/8 exams	High
Poisson (PMF, approx, thinning)	5/8 exams	High
Exponential + memoryless	3/8 exams	Medium
Joint PMF / conditional PMF	3/8 exams	Medium
Covariance / correlation	2/8 exams	Medium
Normal / standardization	2/8 exams	Medium

4-Step Study Plan

Step 1: Copy the Cheat Sheet by Hand

This is itself study. As you write each formula, think about *when* you would use it. The act of transcribing forces you to engage with every formula rather than passively reading. The highlighted formulas on the cheat sheet are the ones that appear on nearly every exam—give them extra attention.

Step 2: Work Every Practice Exam Under Timed Conditions

Give yourself **50 minutes** per exam (matching the real exam length). After each one:

- Check which formulas you needed that you couldn't quickly find on your sheet.
- Rearrange or annotate your hand-written sheet accordingly.
- Grade yourself against the solutions—understand every step you missed.

Priority order for practice exams (most recent = closest to current style):

1. **Spring 2025** and **Fall 2025** — most recent, do these first
2. **Spring 2024** — recent and well-aligned with current curriculum
3. **Fall 2023** and **Fall 2022** — good for additional reps
4. **2021, 2019, 2017** — older but still valuable for pattern recognition

Step 3: Drill These Specific Problem Patterns

These patterns appear almost every year. You should be able to execute each one mechanically:

1. **Bayes with sequential updating.** Given a prior, compute posterior after observing evidence. Then use the posterior as the new prior for the next observation. Appears in medical testing problems, defective-item problems, and similar setups.
   - Key formula: $P(B_j \mid A) = \frac{P(A|B_j)\,P(B_j)}{\sum_i P(A|B_i)\,P(B_i)}$
   - Practice: F2023 Q2, every medical testing variant
2. **Indicator trick for $E[X]$.** Write $X = \sum \mathbf{1}_{A_i}$, then use linearity: $E[X] = \sum P(A_i)$. Works even when the indicators are *dependent*. This sidesteps messy PMF calculations entirely.
   - Appears in: counting problems (expected number of fixed points, matches, etc.)
3. **Geometric series evaluation.** Almost every exam has at least one infinite sum you need to simplify. Know these cold:
   $$\sum_{k=0}^{\infty} r^k = \frac{1}{1-r}, \quad \sum_{k=1}^{\infty} kr^{k-1} = \frac{1}{(1-r)^2}, \quad \sum_{k=0}^{\infty} \frac{\lambda^k}{k!} = e^\lambda$$
4. **First-step analysis.** Condition on what happens on trial 1, write a recursion for the probability or expectation, then solve the resulting equation.
   - Classic example: Gambler's Ruin (2019 Q2c)
   - Template: Let $p_i = P(\text{reach target} \mid \text{start at } i)$. Write p_i in terms of p_{i-1} and p_{i+1}, solve with boundary conditions.
5. **The $E[X^2]$ identity.** $E[X^2] = \mathrm{Var}(X) + (E[X])^2$. The single most useful identity for computing second moments. Use it whenever a problem asks for variance or involves $E[X^2]$.
   - Practice: 2022 Q2b and many variance problems
6. **LOTP decomposition.** When a probability depends on a random quantity N, you **must** condition on N:
   $$P(A) = \sum_n P(A \mid N = n)\,P(N = n)$$
   Similarly for expectations: $E[X] = \sum_n E[X \mid N = n]\,P(N = n)$.

Step 4: Memorize the Common Traps

Grading comments across all past solutions reveal that students lose the most points on these mistakes:

Trap 1: **Assuming independence when it doesn't hold.** If the problem doesn't explicitly state independence, you cannot use $P(A \cap B) = P(A)\,P(B)$ or $\mathrm{Var}(X + Y) = \mathrm{Var}(X) + \mathrm{Var}(Y)$. Always check.

Trap 2: **Not conditioning on random quantities via LOTP.** If a count or parameter is itself random, you must condition on it. Skipping this step is the most common structural error.

Trap 3: **Confusing $E[X^2]$ with $(E[X])^2$.** These are *not* equal unless $\mathrm{Var}(X) = 0$. The difference is exactly the variance.

Trap 4: **Forgetting to divide by P(conditioning event).** When computing $P(A \mid B)$, always divide $P(A \cap B)$ by $P(B)$. Students frequently compute the numerator correctly but forget the denominator.

Trap 5: **Leaving geometric series unsimplified.** The exam expects closed-form answers. Know the series formulas and apply them. Leaving an infinite sum as your final answer will cost points.

Trap 6: **Mutually exclusive $\neq$ independent.** If $A \cap B = \emptyset$ and both have positive probability, they are *dependent*, not independent. This distinction appears in true/false and conceptual questions.

Trap 7: **$P(A \mid B) + P(A \mid B^c) \neq 1$.** The correct identity is $P(A \mid B) + P(A^c \mid B) = 1$. Mixing up the conditioning event vs. the target event is a common error.

Trap 8: **Linearity of expectation needs NO independence; variance of a sum DOES.** $E[X+Y] = E[X] + E[Y]$ always holds. But $\mathrm{Var}(X + Y) = \mathrm{Var}(X) + \mathrm{Var}(Y)$ requires independence (or at minimum uncorrelatedness). In general: $\mathrm{Var}(X + Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) + 2\mathrm{Cov}(X, Y)$.

Exam Day Checklist

- Bring your hand-written cheat sheet (one letter-size page, both sides).
- No calculators, phones, or other aids.
- Read each problem completely before starting to write.
- **Name the distribution** before computing anything—this frames your approach and earns partial credit.
- **State what you're conditioning on** when using LOTP or Bayes—be explicit.
- **Justify each step:** cite the formula, name the theorem, state independence when you use it.
- If stuck, move on and return later. Each problem is worth equal effort.
- Double-check that final answers are simplified (especially geometric series and combinatorial expressions).

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